

Formative Assessment 3 Ramilo

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Case study: Major League Baseball

What is the relationship between payroll and wins among Major League Baseball (MLB) teams? In this homework, we'll find out by wrangling, exploring, and modeling the dataset in `MLPayData_Total.rdata`, which contains the winning records and the payroll data of all 30 MLB teams from 1998 to 2014.

The dataset has the following variables:

- payroll: total team payroll (in billions of dollars) over the 17-year period
- avgwin: the aggregated win percentage over the 17-year period
- Team.name.2014: the name of the team
- p1998, . . . , p2014: payroll for each year (in millions of dollars)
- X1998, . . . , X2014: number of wins for each year
- X1998.pct, . . . , X2014.pct: win percentage for each year

1 Wrangle (35 points for correctness; 5 points for presentation)

1.1 Import (5 points)

- Import the data into a tibble called `mlb_raw` and print it.
- How many rows and columns does the data have?
- Does this match up with the data description given above?

```
# Loading the data set
load("~/RStudioFiles/Data-Wrangling/FA3/ml_pay.rdata")
is_tibble(ml_pay) # Determine if it is a tibble
```

```
## [1] FALSE
```

```
# From a data frame into a tibble
mlb_raw <- as_tibble(ml_pay) # Transform to tibble
is_tibble(mlb_raw)
```

```
## [1] TRUE
```

```
# Print dimensions of the tibble
cat("`mlb_raw` dimensions: ", dim(mlb_raw)[1], "X", dim(mlb_raw)[2])
```

```
## 'mlb_raw' dimensions: 30 X 54
```

- The given data has 30 rows and 54 columns.

- By the description stated above it has data from 1998-2014 which means 17 years of data are involved and amongst those years 3 categories were observed; payroll, number of wins, and win percentage of every team per year which will be 51. Adding the 2 aggregate columns and the team column we would have 54 total columns which is exactly the same as what we have gotten.

1.2 Quality control (15 points)

It's always a good idea to check whether a dataset is internally consistent. In this case, we are given both aggregated and yearly data, so we can check whether these match. To this end, carry out the following steps:

- Create a new tibble called `mlb_aggregate_computed` based on aggregating the data in `mlb_yearly`, containing columns named `team`, `payroll_aggregate_computed`, and `pct_wins_aggregate_computed`.
- Ideally, `mlb_aggregate_computed` would match `mlb_aggregate`. To check whether this is the case, join these two tibbles into `mlb_aggregate_joined` (which should have five columns: `team`, `payroll_aggregate`, `pct_wins_aggregate`, `payroll_aggregate_computed`, and `pct_wins_aggregate_computed`.)
- Create scatter plots of `payroll_aggregate_computed` versus `payroll_aggregate` and `pct_wins_aggregate_computed` versus `pct_wins_aggregate`, including a 45° line in each. Display these scatter plots side by side, and comment on the relationship between the computed and provided aggregate statistics.

Solution.

```
# Split the data set into 2
# Aggregated data
mlb_aggregate <- mlb_raw %>%
  select(Team.name.2014, payroll, avgwin) %>%
  rename(
    team = Team.name.2014,
    payroll_aggregate = payroll,
    pct_wins_aggregate = avgwin
  )

kable(head(mlb_aggregate, 5))
```

team	payroll_aggregate	pct_wins_aggregate
Arizona Diamondbacks	1.120874	0.4902585
Atlanta Braves	1.381712	0.5527605
Baltimore Orioles	1.161212	0.4538250
Boston Red Sox	1.972359	0.5487172
Chicago Cubs	1.459767	0.4736557

```
# year-by-year data
mlb_yearly <- mlb_raw %>%
  select(-c(payroll, avgwin)) %>%
  rename(
    team = Team.name.2014
  ) %>%
  pivot_longer(
    cols = p1998:p2014,
    names_to = "p_year",
    values_to = "payroll"
  ) %>%
```

```

pivot_longer(
  cols = X2014:X1998,
  names_to = "Xyear",
  values_to = "num_wins"
) %>%
pivot_longer(
  cols = X2014.pct:X1998.pct,
  names_to = "Xyear.pct",
  values_to = "pct_wins"
) %>%
mutate(
  p_year = as.numeric(str_remove(p_year, "p")),
  Xyear = as.numeric(str_remove(Xyear, "X")),
  Xyear.pct = as.numeric(str_remove(str_remove(Xyear.pct, "X"), ".pct"))
)

payroll_mlb <- mlb_yearly %>%
  select(team, p_year, payroll) %>%
  distinct(team, p_year, payroll, .keep_all = TRUE) %>%
  rename(
    year = p_year
  )

num_wins_mlb <- mlb_yearly %>%
  select(team, Xyear, num_wins) %>%
  distinct(team, Xyear, num_wins, .keep_all = TRUE) %>%
  rename(
    year = Xyear
  )

win_perc_mlb <- mlb_yearly %>%
  select(team, Xyear.pct, pct_wins) %>%
  distinct(team, Xyear.pct, pct_wins, .keep_all = TRUE) %>%
  rename(
    year = Xyear.pct
  )

mlb_yearly <- payroll_mlb %>%
  left_join(num_wins_mlb, by=c("team", "year")) %>%
  left_join(win_perc_mlb, by=c("team", "year")) %>%
  select(team, year, payroll, pct_wins, num_wins)
kable(head(mlb_yearly, 5))

```

team	year	payroll	pct_wins	num_wins
Arizona Diamondbacks	1998	31.61450	0.3987730	65
Arizona Diamondbacks	1999	70.49600	0.6134969	100
Arizona Diamondbacks	2000	81.02783	0.5246914	85
Arizona Diamondbacks	2001	81.20651	0.5679012	92
Arizona Diamondbacks	2002	102.82000	0.6049383	98

Joining the aggregate data and the computed data

```

# Double checking the aggregated table
mlb_aggregate_computed <- mlb_yearly %>%
  group_by(team) %>%
  summarize(
    payroll_aggregate_computed = sum(payroll)/1000,
    pct_wins_aggregate_computed = sum(num_wins)/(n()*162)
  )

# Joining 2 tables together
mlb_aggregate_joined <- mlb_aggregate %>%
  left_join(mlb_aggregate_computed,by="team")

kable(head(mlb_aggregate_joined,5))

```

team	payroll_aggregate	pct_wins_aggregate	payroll_aggregate_computed	pct_wins_aggregate_computed
Arizona	1.120874	0.4902585	1.222984	0.4901961
Diamondbacks				
Atlanta Braves	1.381712	0.5527605	1.518310	0.5606391
Baltimore Orioles	1.161212	0.4538250	1.305271	0.4538853
Boston Red Sox	1.972359	0.5487172	2.103581	0.5493827
Chicago Cubs	1.459767	0.4736557	1.551726	0.4724038

Creating scatter plots of payroll_aggregate_computed versus payroll_aggregate and pct_wins__ aggregate_computed

```

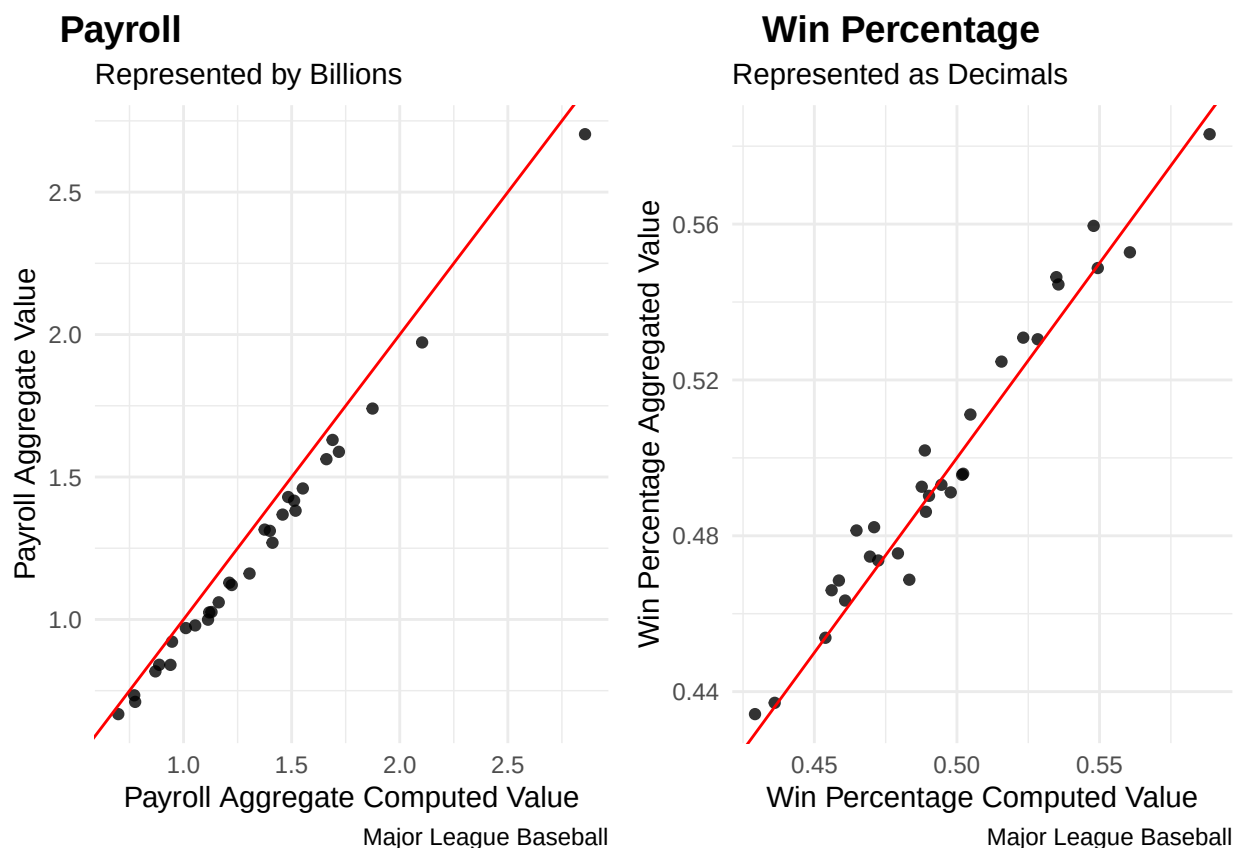
# Creating 2 plots for aggregated vs computed
aggregated_v_computed_plt1 <- ggplot(mlb_aggregate_joined,aes(x=payroll_aggregate_computed,y=payroll_aggregate))
  geom_point(
    size = 1.5,
    alpha = 0.8
  )+
  theme_minimal()+
  labs(
    title = " ",
    caption = "Major League Baseball",
    subtitle = "Represented by Billions",
  )+
  xlab("Payroll Aggregate Computed Value")+
  ylab("Payroll Aggregate Value")+
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "solid", linewidth = 0.5)

aggregated_v_computed_plt2 <- ggplot(mlb_aggregate_joined,aes(x=pct_wins_aggregate_computed,y=pct_wins_aggregate))
  geom_point(
    size = 1.5,
    alpha = 0.8
  )+
  theme_minimal()+
  labs(
    title = " ",
    caption = "Major League Baseball",
    subtitle = "Represented as Decimals",
  )

```

```
)+
xlab("Win Percentage Computed Value")+
ylab("Win Percentage Aggregated Value")+
geom_abline(slope = 1, intercept = 0, color = "red", linetype = "solid", linewidth = 0.5)

# Joining 2 plots together
aggregated_v_computed_plt <- plot_grid(
  aggregated_v_computed_plt1,
  aggregated_v_computed_plt2,
  labels = c("Payroll", "Win Percentage"),
  ncol = 2
)
aggregated_v_computed_plt
```



As shown between the two graphs there is an error within the difference between the aggregated and computed data, however it's error is not that large

2 Explore (50 points for correctness; 10 points for presentation) Now that the data are in tidy format, we can explore them by producing visualizations and summary statistics.

2.1 Payroll across years (15 points)

- Plot payroll as a function of year for each of the 30 teams, faceting the plot by team and adding a red dashed horizontal line for the mean payroll across years of each team.
- Using dplyr, identify the three teams with the greatest payroll_aggregate_computed, and print a table of these teams and their payroll_aggregate_computed.

- Using dplyr, identify the three teams with the greatest percentage increase in payroll from 1998 to 2014 (call it pct_increase), and print a table of these teams along with pct_increase as well as their payroll figures from 1998 and 2014.
- How are the metrics payroll_aggregate_computed and pct_increase reflected in the plot above, and how can we see that the two sets of teams identified above are the top three in terms of these metrics?

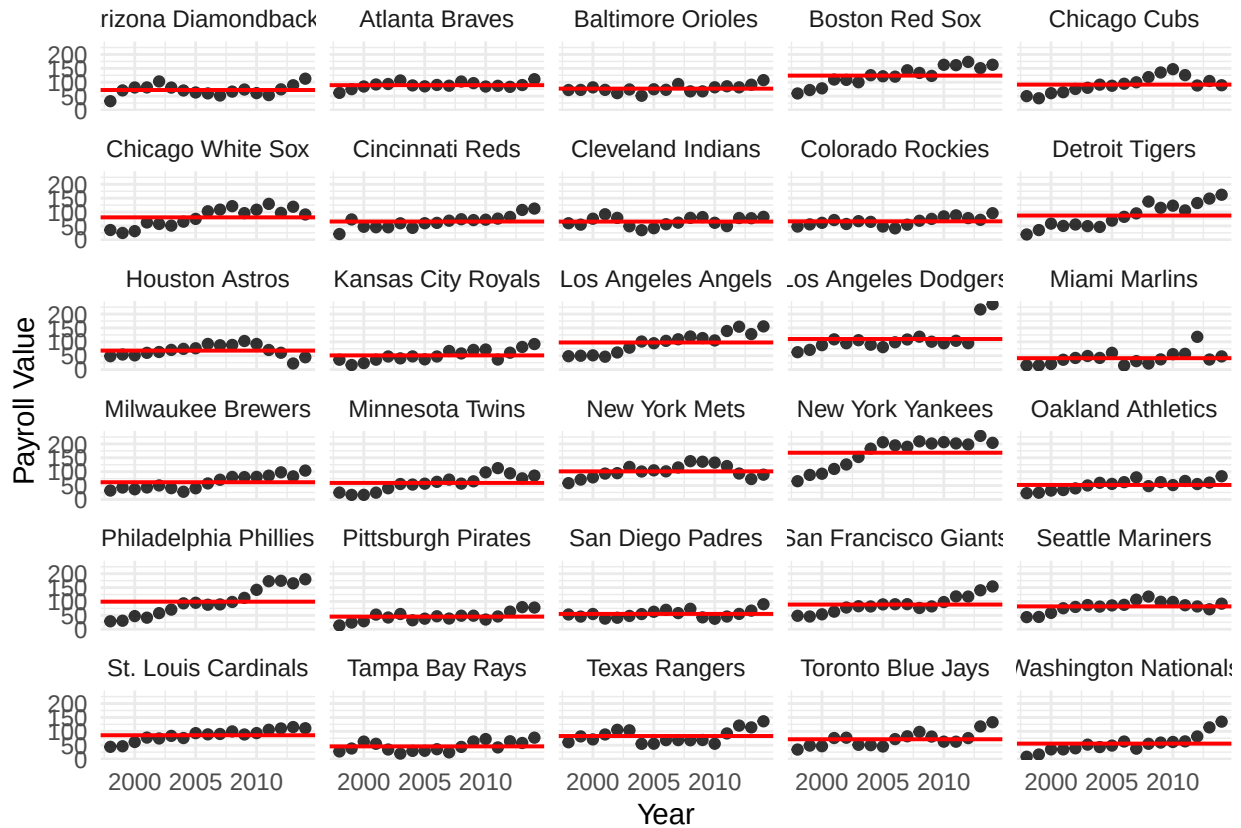
Plotting payroll as a function of year for each team

```
# Plot payroll interms of year
mlb_means <- mlb_yearly %>%
  group_by(team) %>%
  summarize(mean_payroll = mean(payroll, na.rm = TRUE))

payroll_per_year_plt <- ggplot(mlb_yearly, aes(x=year, y=payroll)) +
  geom_point(
    size = 1.5,
    alpha = 0.8
  ) +
  theme_minimal() +
  xlab("Year") +
  ylab("Payroll Value") +
  facet_wrap(~team, ncol=5) +
  geom_hline(data = mlb_means, aes(yintercept = mean_payroll), color = "red", linetype = "solid", size = 1)

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
payroll_per_year_plt
```



Identifying the three teams with the greatest payroll_aggregate_computed

```
# Find highest payroll within the computed data
sorted_payroll_mlb <- mlb_aggregate_computed %>%
  select(team, payroll_aggregate_computed) %>%
  arrange(desc(payroll_aggregate_computed))

kable(head(sorted_payroll_mlb, 3), caption = "Top 3 Highest Payroll")
```

Table 4: Top 3 Highest Payroll

team	payroll_aggregate_computed
New York Yankees	2.857093
Boston Red Sox	2.103581
Los Angeles Dodgers	1.874194

Identifying winrate increase for each team from 1998 and 2014

```
# Find winrate percentage increase
sorted_pct_win_increase <- mlb_yearly %>%
  select(team, year, pct_wins) %>%
  pivot_wider(names_from = "year", values_from = pct_wins) %>%
  mutate(
    pct_increase = `2014` - `1998`
  ) %>%
```

```

left_join(mlb_aggregate_computed,by="team") %>%
select(team,pct_increase,payroll_aggregate_computed) %>%
arrange(desc(pct_increase))
kable(head(sorted_pct_win_increase,3), caption = "Top 3 Highest Win Percentage Increase")

```

Table 5: Top 3 Highest Win Percentage Increase

team	pct_increase	payroll_aggregate_computed
Washington Nationals	0.1575650	0.9466941
Miami Marlins	0.1475849	0.6980929
Detroit Tigers	0.1434805	1.4839380

- Given by the plot above which indicates payroll across the year it is clearly seen that the New York Yankees had their payout increase as the year increase indicating a positive relationship. For the increase of payroll indicated within the plot there is a huge jump from one year to another which signifies a sudden increase in payroll for the teams.

2.2 Win percentage across years (15 points)

- Plot pct_wins as a function of year for each of the 30 teams, faceting the plot by team and adding a red dashed horizontal line for the average pct_wins across years of each team.
- Using dplyr, identify the three teams with the greatest pct_wins_aggregate_computed and print a table of these teams along with pct_wins_aggregate_computed.
- Using dplyr, identify the three teams with the most erratic pct_wins across years (as measured by the standard deviation, call it pct_wins_sd) and print a table of these teams along with pct_wins_sd.
- How are the metrics pct_wins_aggregate_computed and pct_wins_sd reflected in the plot above, and how can we see that the two sets of teams identified above are the top three in terms of these metrics?

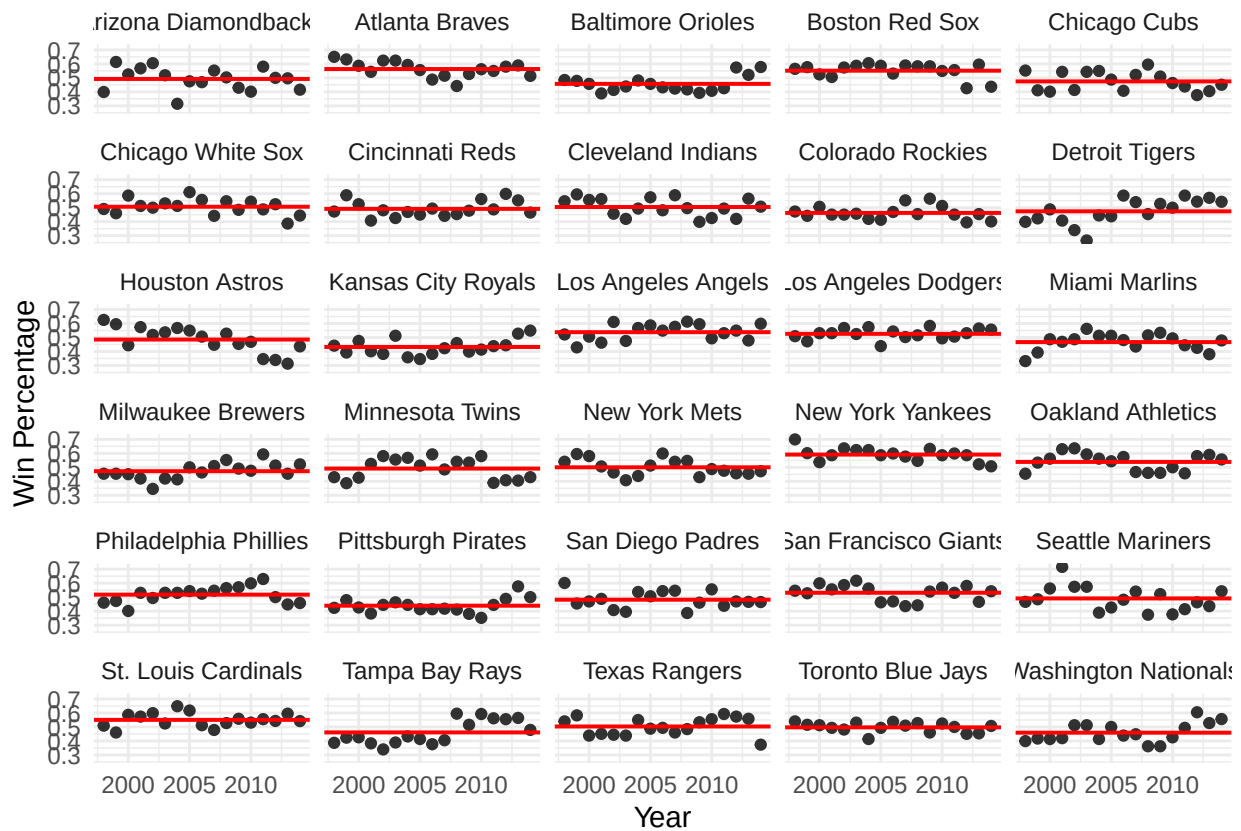
Plot pct_wins as a function of year for each of the 30 teams

```

# Win percentage
mlb_pct_win_means <- mlb_yearly %>%
  group_by(team) %>%
  summarize(mean_pct_win = mean(pct_wins, na.rm = TRUE))

pct_win_per_year_plt <- ggplot(mlb_yearly,aes(x=year,y=pct_wins))+
  geom_point(
    size = 1.5,
    alpha = 0.8
  )+
  theme_minimal()+
  xlab("Year")+
  ylab("Win Percentage")+
  facet_wrap(~team,ncol=5)+
  geom_hline(data = mlb_pct_win_means, aes(yintercept = mean_pct_win), color = "red", linetype = "solid")
pct_win_per_year_plt

```



Identifying highest win percentage

```
# top 3 teams with high pct
sorted_pct_win_mlb <- mlb_aggregate_computed %>%
  select(team,pct_wins_aggregate_computed) %>%
  arrange(desc(pct_wins_aggregate_computed))

kable(head(sorted_pct_win_mlb,3), caption = "Top 3 Highest Win Percentage")
```

Table 6: Top 3 Highest Win Percentage

team	pct_wins_aggregate_computed
New York Yankees	0.5885984
Atlanta Braves	0.5606391
Boston Red Sox	0.5493827

Identifying top 3 most erratic teams

```
# top 3 most erratic teams
sorted_erratic_pct_win_increase <- mlb_yearly %>%
  select(team,pct_wins) %>%
  group_by(team)%>%
  summarise(
    pct_wins_sd = sd(pct_wins)
  ) %>%
```

```
arrange(desc(pct_wins_sd))
kable(head(sorted_erratic_pct_win_increase,3), caption = "Top 3 Most Erratic Teams")
```

Table 7: Top 3 Most Erratic Teams

team	pct_wins_sd
Houston Astros	0.0914336
Detroit Tigers	0.0897654
Seattle Mariners	0.0892271

- Through the plot above it is shown that the teams with the top highest win percentage have consistently been able to perform good through out the progression of the game. Most erratic teams have points that mostly don't align with the mean of each win percentage, to which means that throughout the year this team performs significantly differently for each iteration.

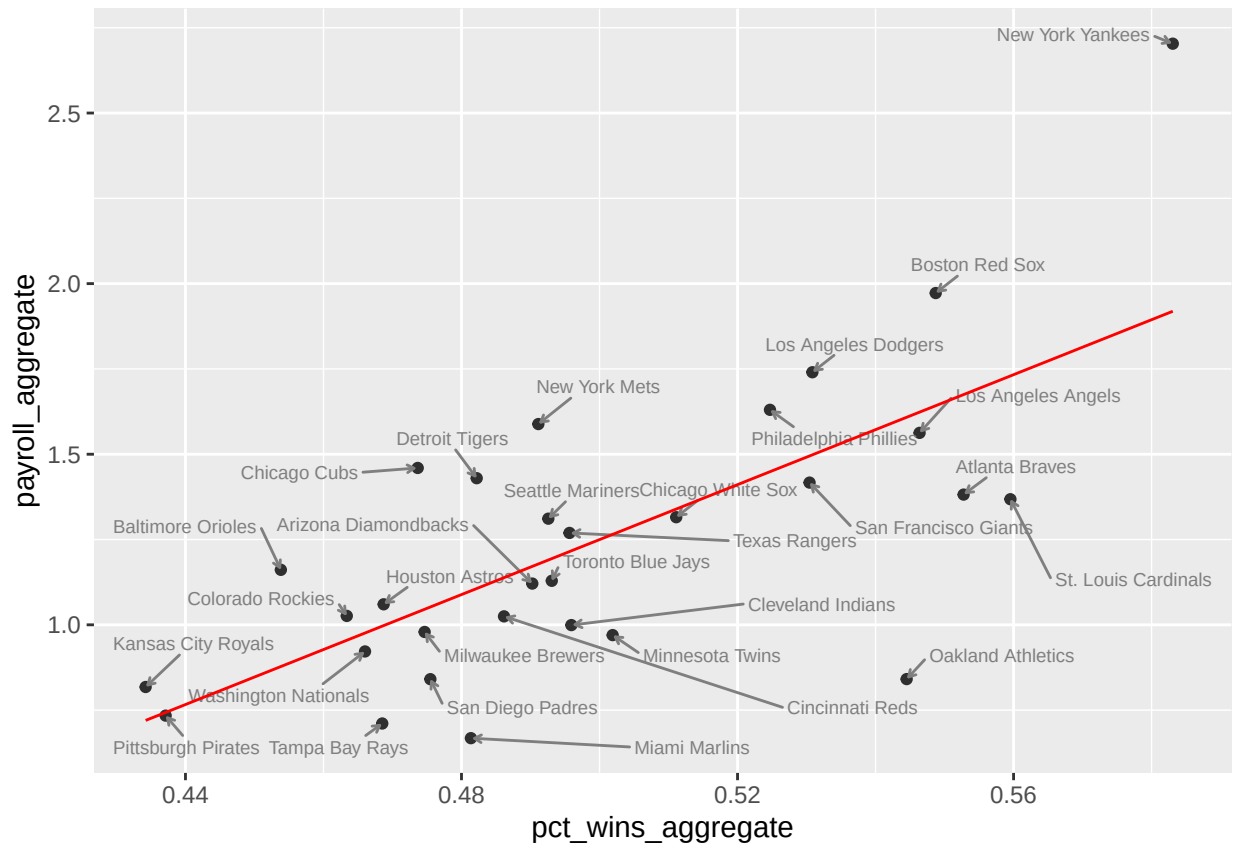
2.3 Win percentage versus payroll (15 points)

Let us investigate the relationship between win percentage and payroll.

- Create a scatter plot of pct_wins versus payroll based on the aggregated data, labeling each point with the team name using geom_text_repel from the ggrepel package and adding the least squares line.
- Is the relationship between payroll and pct_wins positive or negative? Is this what you would expect, and why?

```
# Plot win percentage and payroll
win_perc_v_payroll <- ggplot(mlb_aggregate, aes(x = pct_wins_aggregate, y = payroll_aggregate)) +
  geom_point(
    size = 1.5,
    alpha = 0.8
  ) +
  geom_text_repel(
    aes(label = team),
    size = 2.5,
    min.segment.length = 0,
    seed = 42,
    box.padding = 0.5,
    max.overlaps = Inf,
    arrow = arrow(length = unit(0.010, "npc")),
    nudge_x = .001,
    nudge_y = .001,
    color = "grey50"
  ) +
  geom_smooth(method = "lm", se = FALSE, color = "red", size = 0.5)
win_perc_v_payroll
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



- The relationship indicated by the plot shows that there exist a positive relationship between payroll and the win percentage of the team, meaning as the payroll increases, it is highly likely that it is due to the win percent increasing. This is to be expected since if the team improves their performance their value as players and as a team increases thus increasing their payrolls.

2.4 Team efficiency (5 points)

Define a team's efficiency as the ratio of the aggregate win percentage to the aggregate payroll—more efficient teams are those that win more with less money.

- Using dplyr, identify the three teams with the greatest efficiency, and print a table of these teams along with their efficiency, as well as their `pct_wins_aggregate_computed` and `payroll_aggregate_computed`.
- In what sense do these three teams appear efficient in the previous plot?

```
# Identify the most efficient teams
mlb_aggregate_computed <- mlb_aggregate_computed %>%
  mutate(
    team_efficiency = pct_wins_aggregate_computed / payroll_aggregate_computed
  ) %>%
  arrange(desc(team_efficiency))

kable(head(mlb_aggregate_computed,3), caption = "Top 3 Most Efficient Teams")
```

Table 8: Top 3 Most Efficient Teams

team	payroll_aggregate_computed	pct_wins_aggregate_computed	team_efficiency
Miami Marlins	0.6980929	0.4647785	0.6657832
Oakland Athletics	0.8875729	0.5355846	0.6034261
Tampa Bay Rays	0.7761869	0.4586057	0.5908444

- The ratio is describing the efficiency of each teams where we want to determine the teams that has a high percentage but a relatively low payroll, implying the identified teams are highly efficient because of it.